QuarryChain

Advanced Blockchain for Business

WHITEPAPER VERSION 2.0

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00 Introduction

QuarryChain is an innovative blockchain network designed to revolutionize how developers create and deploy smart contracts and decentralized applications (DApps). Our platform offers an ecosystem of tools that empower users to freely own and store data at high speed, low cost, and with minimal environmental impact.

What is a Blockchain?

Blockchain(Distributed Ledger Technology) is a Database that is consensually shared, replicated, and synchronized.

To better understand this, let's say for example, you have an individual who decides to deposit an amount of money into a banking institution, usually the individual trusts that the amount will be there until they decide to exchange it for goods and services. This individual trusts that the Bank will have an accurate record of the transaction, such as the amount, depositor, date, and time of the deposit.

Society today constantly relies on central repositories, such as Banks or Governments, to collect, maintain, and protect the recorded actions of individuals or institutions. Blockchain however, changes this game completely. It's difference from a centralized repository is that it completely decentralizes the source of trust.

For this to work, an individual will deposit funds into a digital wallet and the value will be captured on the Blockchain. For example, let's say an individual purchased a digital asset, the transaction would then be captured in the Blockchain along with the charge at the fund level in the Digital Account. The bank is not required as a Trusted Third Party. The Trustworthy record is then recorded in the Blockchain shared by all the parties on the network.

Technical Description

The replication and storage of transactional data by each party, or node, on a Blockchain Network is known as a Distributed Ledger. Conflicts, or inaccuracies within the Database, are automatically resolved with predefined ledger rules. The fundamental characteristics of the distributed ledger include:

- Operation with Peer-to-Peer Networks
- Decentralized transaction record keeping
- Consensus or trust-based Transactions
- Tamper Resistance

Blockchains, while similar to Databases, are not used for general data storage, but rather instead hold information about Transactions. Sometimes the Blockchain will contain the Transactions themselves or may include the Proof of Transactions is Valid.

The Three Core Foundations of Blockchain

BLOCK: Block: A list of recorded transactions over a period of time. Transactions can represent virtually any type of activity, from registering a land deed to a single purchase. Any rules relating to the block itself are established when the Network is first created.

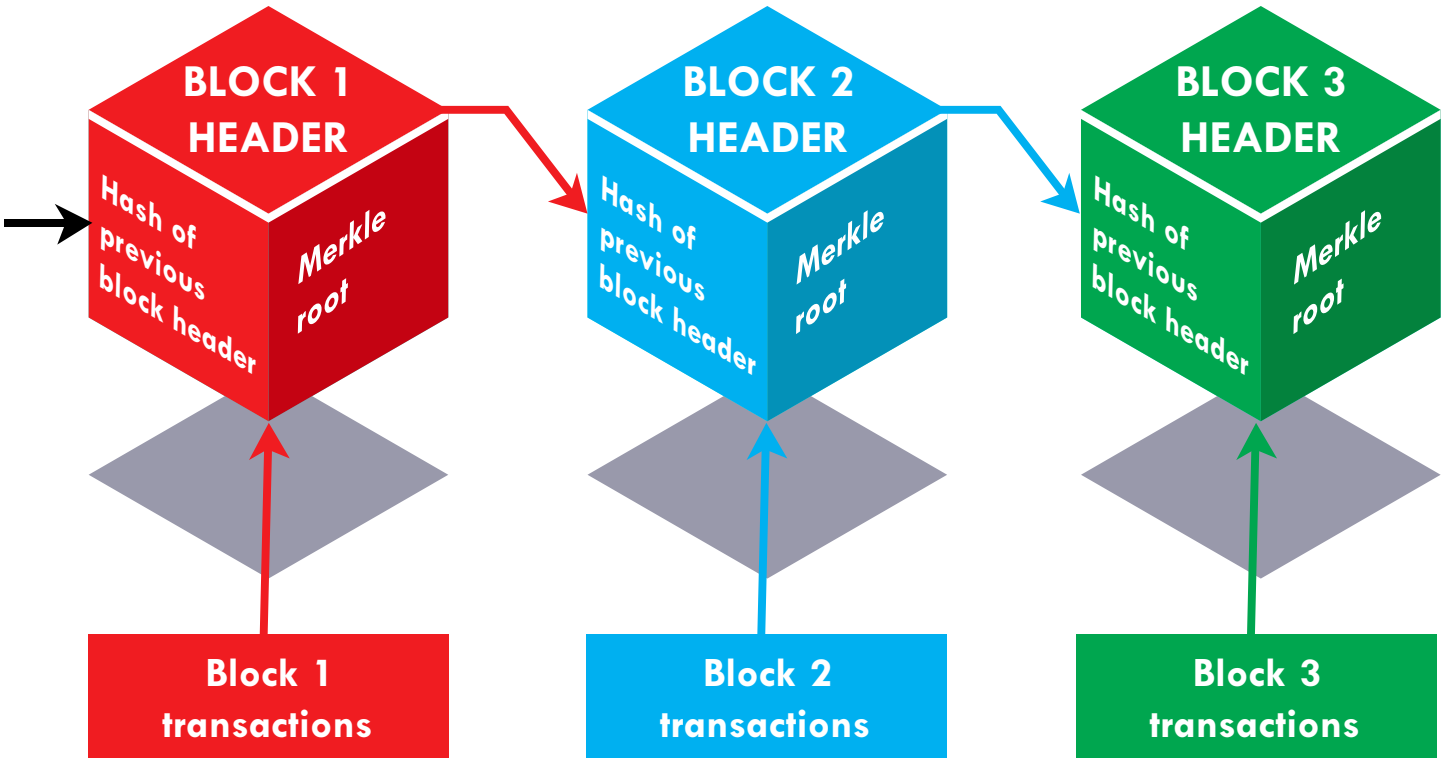
CHAIN: Chain: When the Block reaches its maximum size of transactions, it is chained or linked to the proceeding block through a hash(described in the section below) The hash value of one Block is inserted into the next Block. This makes a link between the new Block and the previous Block. Repeating a hash function on an unaltered block of data will always generate the same fixed length value. If a Block of Data is altered, the resulting hash output will be different. A user can then see the hashes are different and will know the original Block has been altered and may no longer be trustworthy.

NETWORK: Network: The Network is made up of nodes each containing a complete record of all transactions on a blockchain. No centralized "official" copy exists and no node is "trusted" more than another. The Data integrity is maintained by the blockchain being replicated on all of the nodes. The best way to think of a Node would be to think of it as a cluster of servers running on a Blockchain. Node operators are incentivized to operate on a node by receiving rewards for their efforts.

With Cryptocurrencies for example, nodes compete to solve Crypto-Puzzles. The first node completing the puzzle has its solution verified by other nodes. Once the solution is verified, the node completing the puzzle adds the next block to the blockchain and is also rewarded with Cryptocurrency for its effort. This process is called Mining, with the resources involved called Miners. Nodes are found all across the globe and are challenging to operate. For example, the infrastructure of one Cryptocurrency is supported by approximately 5000 nodes.

Each Blockchain has its own rules or algorithms governing how nodes validate transactions intended for entry into the blockchain. These are known as a Consensus Mechanism, and it is established when the blockchain has been created. By integrating a Consensus Mechanism, blockchains create a way for parties who do not know if they can trust each other to agree an entry should be added to the blockchain. This confronts the Byzantine Generals Problem. Each blockchain will have its own consensus mechanism depending on the type of transaction it is capturing. Some of these mechanisms are known as "proof of work", "proof of space" or "proof of stake". The most important thing to note is that these mechanisms facilitate authenticity, or the immutability of transaction records.

With blockchain technology, each page in a ledger of transactions forms a block. That block has an impact on the next block or page through cryptographic hashing. In other words, when a block is completed, it creates a unique secure code, which ties into the next page or block, creating a chain of blocks, or blockchain.



SIMPLIFIED BITCOIN BLOCKCHAIN

What is a Hash?

A hash is an algorithm that takes a variable string data and generates a fixed length value. The data about the transactions must be small so the validity of the transaction can be quickly calculated and distributed to other nodes.

Large amounts of data are often stored "off chain" with pointers or hashes of the data stored within the blockchain. Traditionally, ledgers were used to record transactions of property or goods: only the transaction was captured in the ledger with the real property being handled separately.

Let's say for example, a Secure Hash Algorithm generating a 256-bit signature (SHA-256) generates different hash values for slight variations in the spelling of the words "National Archives".

Variable Text String Entry Value	Hash Value
National Archives	6429799B9AF2D91CBF915CB0290F3A50281193A977B-3457D63E4541 CC5788C5B
National Archives (an extra space)	0CF085A8AD9B352925790548D34FBD9569D32C160963061D-066CD4 C24481AC9B
Nati0nal Archives	5F2D570FC940D5F8DE89310DB43F789FDD99F51E-89C021E1A50ACB7 A6FE2CF83

Usually, to save space, there will be multiple hash values that can be brought together and hashed again, creating a single hash value or Merkle root that represents multiple hashes. This technology is known as the Merkle Tree.

NARA (National Archives and Records Administration) began hashing electronic records when the Electronic Records Archive (ERA) was deployed in 2008. NARA has been using the SHA-256 algorithm to prove file fixity when electronic records are ingested into ERA. The files are then automatically hashed and this can be used to verify the authenticity of a file. The hash is then stored in log files and the processing archivist has access to this information, usually through reports. In the future however, it might be more feasible to store this information in a blockchain.

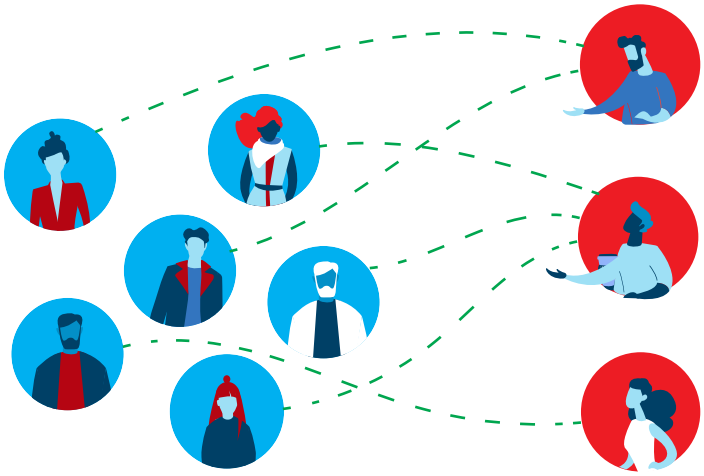
01 QuarryChain Architecture

The QuarryChain Network adopts a 3-layer architecture divided into a Storage Layer, Core Layer, and Application Layer. The QuarryChain Protocol adheres to Google Protobuf, which intrinsically supports multi-language extension.

1.1 Core

There are several modules in the core layer, including smart contracts, account management, and consensus. A stack-based virtual machine is implemented on QuarryChain and an optimized instruction set is used. In order to better support DApp developers, Solidity was chosen as the smart contract 4 language, followed by future support of other advanced languages. In addition, QuarryChain's consensus mechanism is based on Delegated Proof of Stake (DPoS) and many innovations were made in order to meet its unique requirements.

Nodes express Interest to become a block contributor/witness by staking some of their tokens and also through lobbying in the community.



People in the network allocate some of their tokens as votes for witnesses. The more tokens they have, the higher their voting weight, hence the *proof of stake*. They do not give the tokens to the witness, but rather can reallocate them as expression of choice and may allocate them to another witness if they change their mind.

A function will randomly choose a witness from the nodes with the higher weight of votes.

The chosen node will add their block to the chain for validation.

The other witnesses in the network will validate the block for confirmation. if the block is validated, the witness is rewarded the block transaction fee.

1.2 Storage

The QuarryChain Network designed a unique distributed storage protocol consisting of Block Storage and State Storage. The notion of a graph database was introduced into the design of the storage layer to better meet the need for diversified data storage in the real world.

1.2.1 Blockchain Storage

QuarryChain's blockchain storage chooses to use LevelDB, which is developed by Google and proven successful with many companies and projects. It has high performance and supports arbitrary byte arrays as both keys and values, singular get, put and delete, batched put and delete, bi-directional iterators, and simple compression using the very fast Snappy algorithm.

1.2.2 State Storage

The QuarryChain Network has a KhaosDB in the full-node memory that can store all the newly forked chains generated within a certain period of time and supports witnesses to switch from their own active chain swiftly into a new main chain. It also can protect blockchain storage by making it more stable from being terminated abnormally in an intermediate state.

1.3 Application

Developers can create a diverse range of DApps and customized wallets on the QuarryChain Network. Since the Network enables smart contracts to be deployed and executed, the opportunities of unity applications are unlimited.

1.4 Protocol

The QuarryChain Protocol adheres to Google Protocol Buffers, which is a language-neutral, platform-neutral and extensible way of serializing structure data for use in communications protocols, data storage, and more.

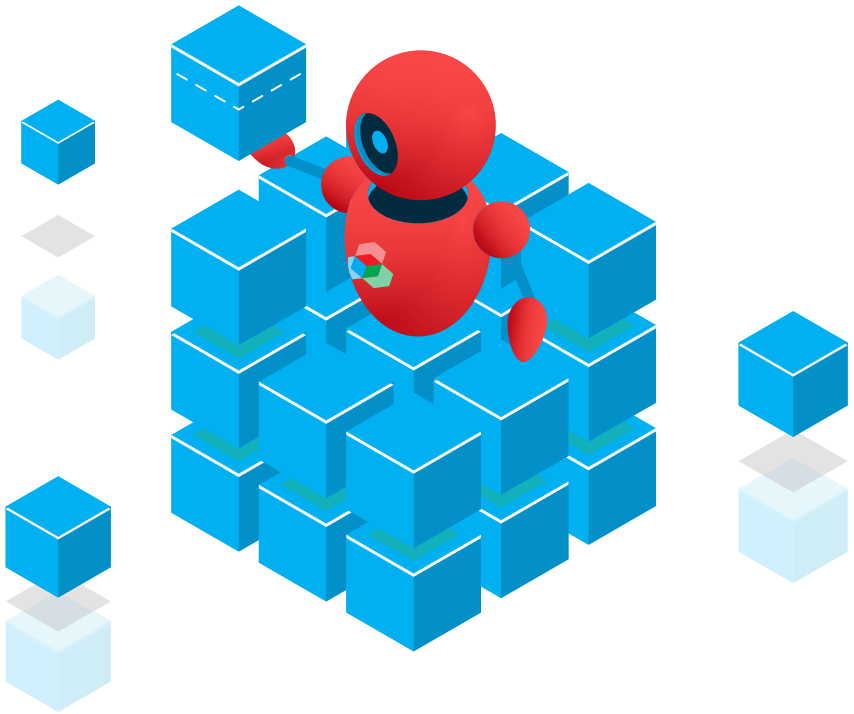
1.4.1 Protocol Buffers

Protocol Buffers (Protobuf) is a flexible, efficient, automated mechanism for serializing structured data, similar to JSON or XML, but much smaller, faster and simpler. Protobuf (.proto) definitions can be used to generate code for C++, Java, C#, Python, Ruby, Golang, and Objective-C languages through the official code generators. Various third-party implementations are also available for many other languages. Protobuf eases development for clients by unifying the API definitions and also optimizing data transfers. Clients can take the API .proto from QuarryChain's protocol repository and integrate through the automatically-generated code libraries

As a comparison, Protocol Buffers are 3 to 10 times smaller and 20 to 100 times faster than XML, with less ambiguous syntax. Protobuf generates data access classes that are easier to use programmatically.

1.4.2 HTTP

The QuarryChain Protocol provides a RESTful HTTP APO alternative to the Protobuf API. They all share the same interface but the HTTP API can be readily used in JavaScript clients.



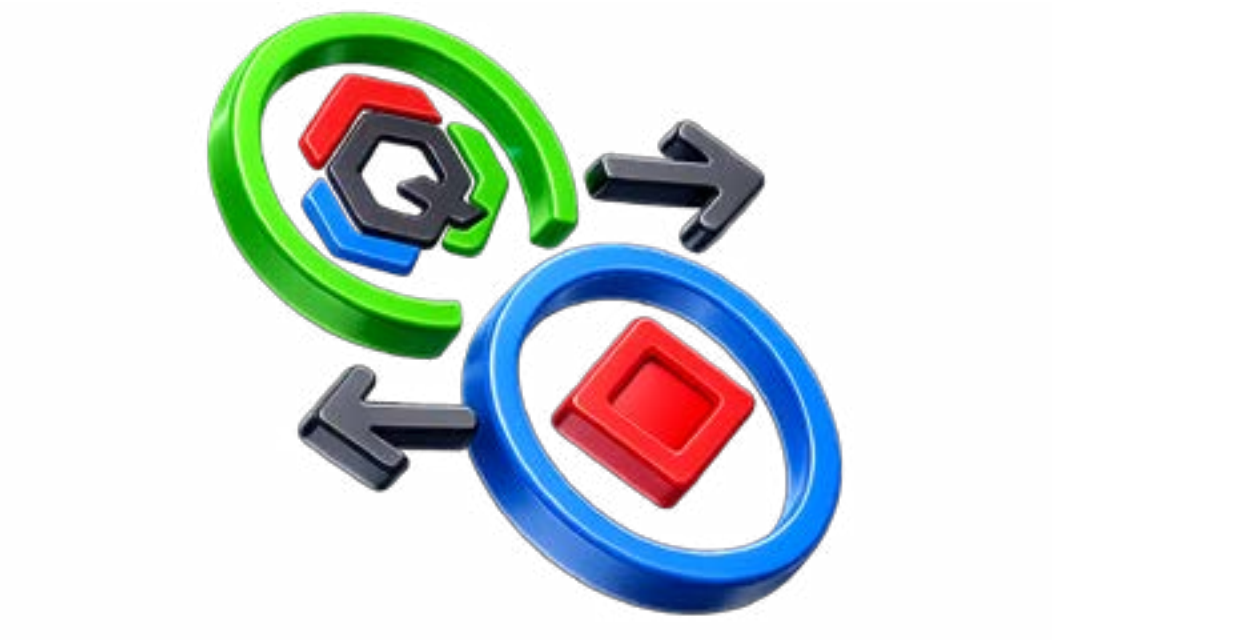
1.5 QuarryChain Virtual Machine (QVM)

The QVM is a lightweight, Turing complete virtual machine developed for QuarryChain's Ecosystem. The QVM connects seamlessly with the existing development ecosystem to provide millions of global developers with a custom-built blockchain system that is efficient, convenient, stable, secure, and scalable.

1.6 Decentralized Exchange (DEX)

The QuarryChain Network natively supports decentralized exchange functions. A decentralized exchange consists of multiple trading pairs. A trading pair (notation "Exchange") is an Exchange Market between QL.

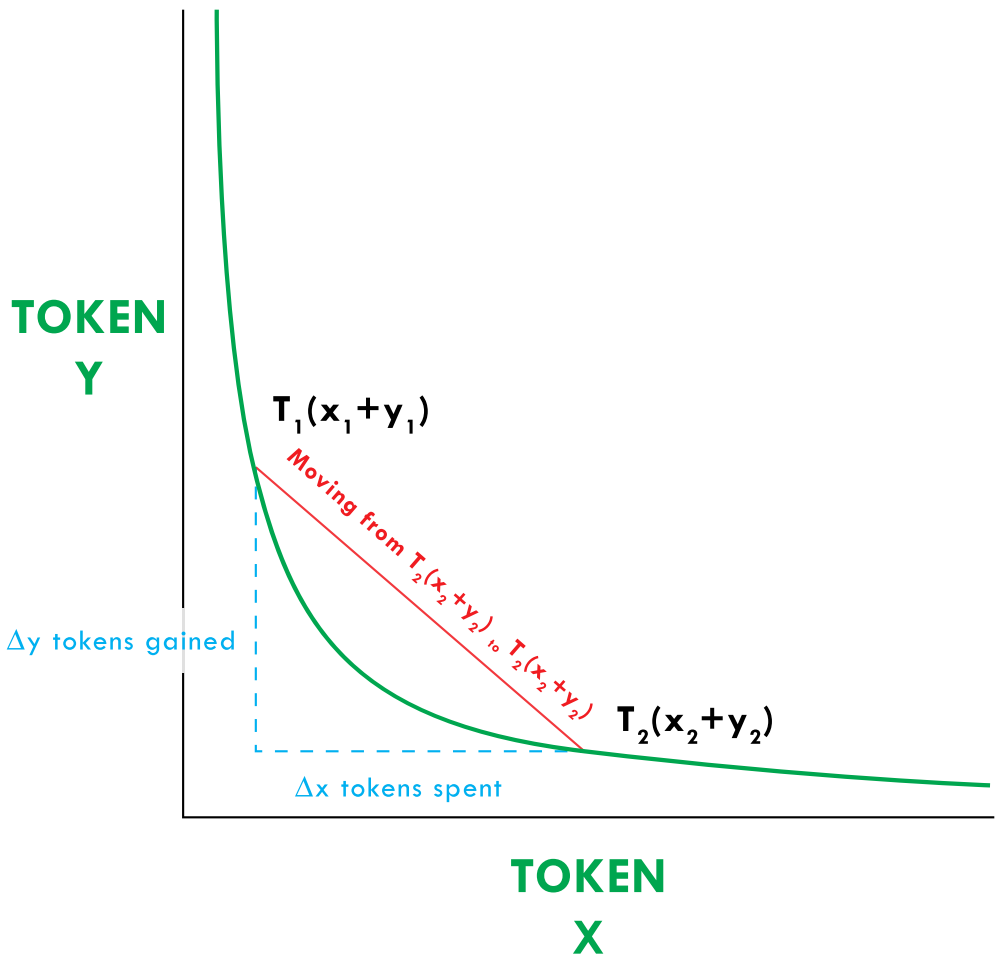
What is QuarrySwap?



QuarrySwap is a decentralized exchange (DEX, for short), and one of QuarryChain's first Applications. QuarrySwap also happens to be non-custodial, meaning that unlike centralized exchanges, Quarryswap does not need to possess your tokens in order for you to be able to trade them. Instead, QuarrySwap allows users to trade trustfully, peer-to-peer, with liquidity that is supplied by other users. This means that new projects can easily connect to their desired markets as long as some entity is willing to provide the liquidity.

In order to be a Liquidity Provider, holders of any token need to supply equal parts liquidity for that token (sometimes known as the quote token), and a second token (usually ETH, or a stable coin). In return, these holders receive QuarrySwap liquidity provider tokens (QLP) that represent their share of the pooled liquidity for that token pair. The existence of this pooled liquidity gives other traders access to the underlying tokens in exchange for a small fee, which is distributed proportionately to all the Liquidity Providers.

In a way, QuarrySwap also acts as an "automated market maker" (or AMM, for short). While a user's underlying tokens remain in the pool, fluctuations in the price of the two underlying tokens automatically recalibrate the quantity of those tokens to conform the equations x*y=k, where x and y are the quantities of the two paired tokens, and k is constant. This means that even though you supply equal parts of the two tokens to the pool, but also the quantities you receive when you reclaim your liquidity will change relative to the difference in the change in the proof of the two tokens when you remove the liquidity. If the price of x token goes up, and y token goes down, you will have less of x and more of y, and vice versa. If the price of both tokens goes up, or the price of both goes down, you will nonetheless have relative quantities of each token proportionately to the difference in the change of the pierce of x and y.



What are Liquidity Pools?

Liquidity Pools are otherwise known as a place to pool tokens (which is also known as Liquidity) so that users can use them to make trades in a decentralized way. These pools are created by users and decentralized apps (Dapps for short) who want to profit from their usage. To pool liquidity, the amounts a user supplies must be equally divided between two coins:

- The Primary Token (QRY)
- Base Token (ETH/BNB)

QuarrySwap's liquidity pools will allow anyone to provide liquidity through the exchange, and when they do so, they will receive QLP tokens (QuarrySwap Liquidity Provider Tokens). If a user happens to deposit $QRC and $ETH into a pool, they would receive QRC-ETH QLP tokens. These tokens represent a proportional share of the pooled assets, allowing a user to reclaim their funds at any point. Every time another user uses the pool to trade between $QRC and $ETH, a 0.3% fee is taken on the trade. 0.25% of the rest goes back to the LP Pool.

The value of the QLP tokens, which represent the shares of the total liquidity in each pool, is then updated with each trade to add their value relative to the tokens the pool uses to trade. If previously there were 100 QLP Tokens representing 100 ETH and 100 QRC, each token would be worth 1 ETH & 1 QRC(note in this example, ETH and QRC are the same relative value). If a user were to trade 10 ETH for 10 QRC in that pool, and another user were to trade 10 QRC for 10 ETH, then there would now be 100.025 ETH and 100.025 QRC. This means that each LP token would be worth 1.0025 ETH and 1.00025 QRC now when it is withdrawn.

Instructions for Adding Liquidity

1. 1. Adding liquidity is never really straightforward. First, you need to choose which tokens to input. Click "Select Token"
2. 2. If you have never used QuarrySwap before, this will trigger a popup asking you to choose a list of tokens. Click on "Choose a list".
3. 3. There are different lists to choose from, but in this example we'll be using the QuarrySwap Menu.
4. 4. After choosing a list, you will be shown the tokens represented in that list along with your balance of them (if any). In this case, we only have two to choose from, QRY and ETH.
5. 5. After selecting our desired input tokens, we're ready to click "Supply"! If this is your first time interacting with QuarrySwap's smart contracts for either or both of the tokens in question,

you will be prompted to "Approve" that our smart contracts are allowed to handle them for you.

6. 6. Once you've added the Liquidity, you will receive QuarryChain Liquidity Provider Tokens (QLP) in return. These represent your share in the liquidity pool, and your earned fees that are received upon the withdrawal of the liquidity.

What is Yield Farming?

When people say that they are Yield "farming", they are actually talking about giving something to a platform(usually by staking their assets) that justifies paying out an annualized percentage yield (APY). Often this effort incentivizes adding liquidity to a network, and then staking liquidity provider tokens, or adding liquidity to a market for borrowing and lending. Projects will incentivize this behavior when it enables them to function more efficiently. QuarryChain has by and large the most effective DeFi for farming yield. This is in part because nearly every token on QuarryChain is natively yield bearing, which enables QuarryChain to function very efficiently. In the case for QuarryChain's Ecosystem, adding liquidity to any branch of the platform allows the platform as a whole to be more efficient, and every type of market is available on the QuarryChain Network. It has its own exchange(QuarrySwap), and it has staking derivatives. The QuarryChain Network is able to deliver a compounded yield farming experience by "stacking" these three yields in simple procedures.

How do I stack Yield?

There's three types of yield on the QuarryChain Network (and arguably three types of yield in DeFi). The QuarryChain Network provides a place where you can effectively stack these three yield types on top of each other. To understand the full scope of the Network, let;s try to understand how to stack these yields.

Here is a description of the three types of Yield.

PROTOCOL USAGE

Whenever someone makes a trade on QuarrySwap, 0.25% swap fee's are distributed among liquidity providers, and 0.05% of all swaps on all chains are accrued by QuarryChain's Ecosystem Token Holders.

NETWORK TOKENS

QLP tokens are given rewards, and these rewards further optimize your impermanent gain (yields earned through volume and volatility).

LEVERAGE DEMAND

This can be expressed as the interest rate for lenders on QuarrySwap, demand for leverage in crypto markets drives increased borrowing.

Simple Stacking: 2 Types of Yield

Now that you understand what the three forms of yield are, you can try to stack them yourself. Start small(if you like) by finding a token pair that you want to hold, and using it to stack two types of yield on QuarrySwap. Usually, the pairs require Ether and a second token, but the following example can be any pair that's on that's on the Onsen menu, or the permanent menu:

1. Provide liquidity on QuarrySwap by entering a pool that is incentivized by QuarryChain Tokens.
2. Deposit the Liquidity Token (QLP) you received after supplying liquidity into a yield farm, and earn $QUARRY.

Full Stack: 3 Types of Yield

If you think you're ready to try something a little more complicated, you can try stacking three types of yields as described below. As you can see, stacking three types of yields requires utilitizing TUFF.

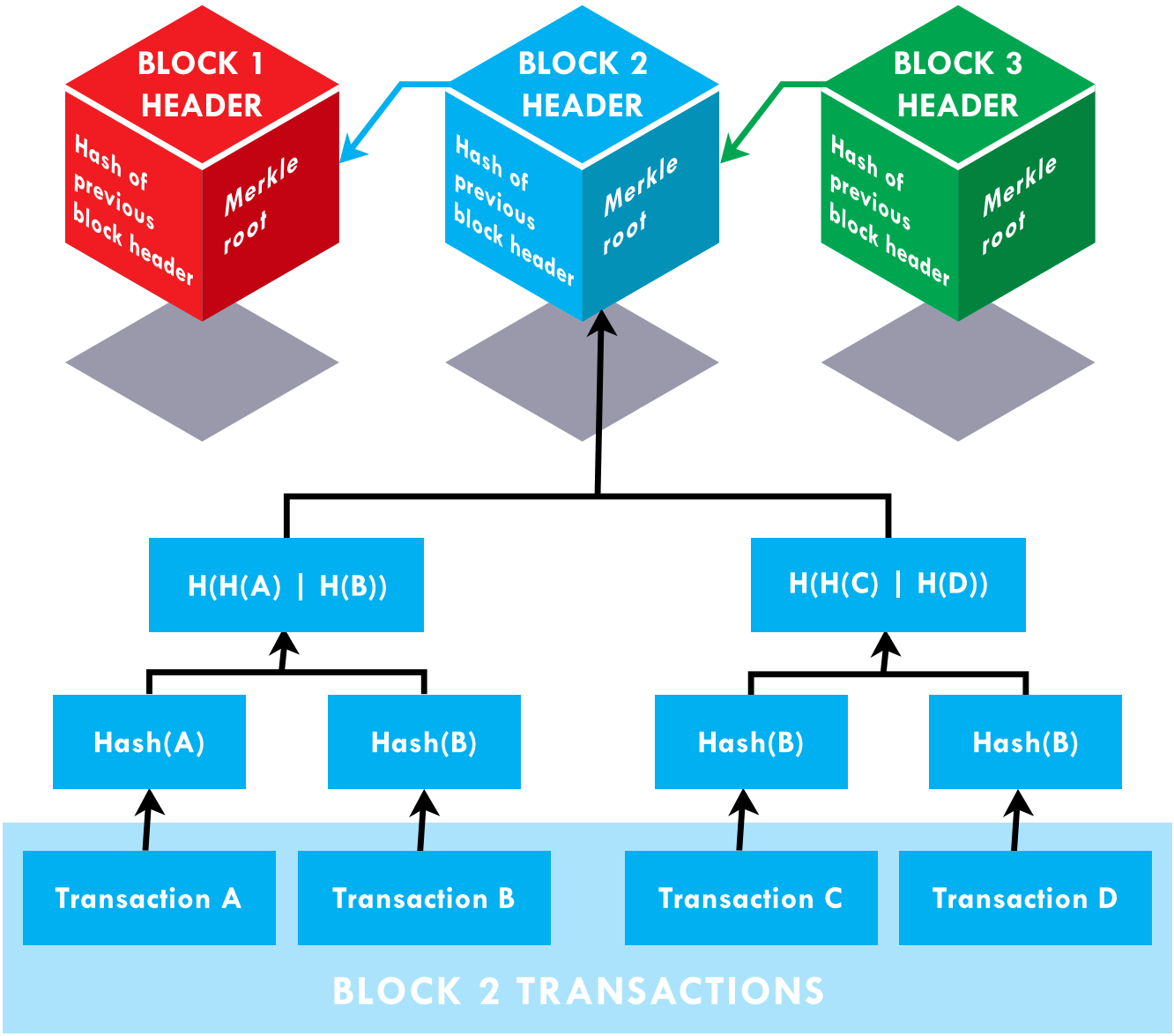
1. Lend an asset to the QuarrySwap Market, and receive a (QLP) token in return.
2. Take the (QLP) token and stake into a yield farm, to earn QuarryChain Tokens.
3. Stake QuarryChain Token Rewards for additional yield in the QuarryChain Ecosystem

02

QuarryChain Consensus Mechanism

2.1 Delegated Proof of Stake

The earliest consensus mechanism is the Proof of Work (PoW) consensus mechanism. This protocol is currently implemented in Bitcoin and Ethereum . In PoW systems, transactions broadcast through the network are grouped together into nascent blocks for miner confirmation. The confirmation process involves hashing transactions using cryptographic hashing algorithms until a merkle root has been reached, creating a **merkle tree:**



Cryptographic hashing algorithms are useful in network attack prevention because they possess several properties.

INPUT/OUTPUT SIZE:	The algorithm can pass in an input of any length in size, and outputs a fixed length hash value.
EFFICIENCY:	The algorithm is relatively easy and fast to compute.
PRE-IMAGE RESISTANCE:	For a given output z, it is impossible to find any input x such that h(x) = z. In other words, the hashing algorithm h(x) is a one-way function in which only the output can be found, given an input. The reverse is not possible.
COLLISION RESISTANCE:	It is computationally infeasible to find any pairs x1 ≠ x2 such that h(x1) = h(x2). In other words, the probability of finding two different inputs hashing to the same output is extremely low. This property also implies second pre-image resistance.
SECOND PRE-IMAGE RESISTANCE:	Given x1 , and thus h(x1), it is computationally infeasible to find any x2 such that h(x1) = h(x2). While this property is similar to collision resistance, the property differs in that it is saying an attacker with a given x1 will find it computationally infeasible to find any x2 hashing to the same output.
DETERMINISTIC:	Maps each input to one and only one output.
AVALANCHE EFFECT:	A small change in the input results in an entirely different output.

These properties give the cryptocurrency network its intrinsic value by ensuring attacks do not compromise the network. When miners confirm a block, they are rewarded tokens as a built-in incentive for network participation. However, as the global cryptocurrency market capitalization steadily increased, the miners became centralized and focused their computing resources on hoarding tokens as assets, rather than for network participation purposes. CPU miners gave way to GPUs, which in turn gave way to powerful ASICs. In one notable study, the total power consumption of Bitcoin mining has been estimated to be as high as 3 GW, comparable to Ireland's power consumption. This same study projected total power consumption to reach 8 GW in the near future.

To solve the energy waste issue, the Proof of Stake (PoS) consensus mechanism was proposed by many new networks. In PoS networks, token holders lock their token balances to become block validators. The validators take turns proposing and voting on the next block. However, the problem with standard PoS is that validator influence correlates directly to the amount of tokens locked up. This results in parties hoarding large amounts of the network's base currency wielding undue influence in the network ecosystem.

Proof of Work VS Proof of Stake

The probability of mining a block is dependent on how much work/computing is done by the miner. Highly competitive.

The mining and market can be monopolized or controlled by anyone with 51% of the computing power.

The first miner to solve the block equation eceives a reward for their work.

The block creator is chosen by the user stake algorithm. A person can 'mine' depending on how many coins they hold.

Proof of Stake system makes any 51% hacking or controlling tactics more expensive to carry out.

There is no reward for making a block, instead the creator takes a transaction fee.

The QuarryChain consensus mechanism uses an innovative Delegated Proof of Stake system in which 27 Quarry Miner (QM) produce blocks for the network. Every 6 hours, QuarryChain Token account holders who freeze their accounts can vote for a selection of QM candidates, with the top 27 candidates deemed the QM. Voters may choose QM based on criteria such as projects sponsored by QM to increase QuarryChain's Token adoptions, and rewards distributed to voters. This allows for a more democratized and decentralized ecosystem. QM' accounts are normal accounts, but their accumulation of votes allows them to produce blocks. With the low throughput rates of Bitcoin and Ethereum due to their PoW consensus mechanism and scalability issues, QuarryChain's DPoS system offers an innovative mechanism resulting in 20,000 - 100,000 TPS compared to Bitcoin's 3 TPS and Ethereum's 15 TPS.

The QuarryChain protocol network generates one block every three seconds, with each block awarding 32 Quarry Tokens to Quarry Miner. A total of 336,000 Quarry Tokens will be awarded annually to the 27 QM. Each time an QM finishes block production, rewards are sent to a sub-account in the super-ledger. QM can check, but not directly make use of these Quarry tokens. A withdrawal can be made by each QM once every 24 hours, transferring the rewards from the sub-account to the specified QM account.

The three types of nodes on the QuarryChain network are Witness Node, Full Node, and Solidity Node. Witness nodes are set up by QM and are mainly responsible for block production and proposal creation/ voting. Full nodes provide APIs and broadcast transactions and blocks. Solidity nodes sync blocks from other Full Nodes and also provide indexable APIs

03

QuarryChain Account

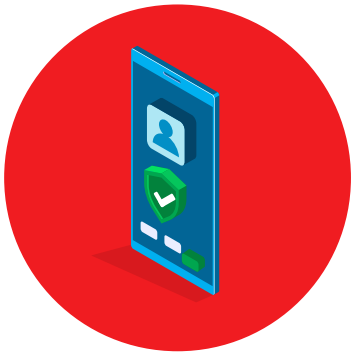
3.1 Types

The three types of accounts in the QuarryChain Network are regular accounts, token accounts, and contract accounts.

1. Regular accounts are used for standard transactions.
2. Token accounts are used for storing QRC-10 Tokens.
3. Contract accounts are smart contract accounts created by regular accounts and can be triggered by regular accounts as well.

3.2 Creation

There are three ways to create a QuarryChain Account:



1
Create a new account through using the API

2
Transfer QRC into a new account address

3
Transfer any QRC-10 into a new account address

An offline key-pair consisting of an address (public key) and a private key, and not recorded by the QuarryChain Network, can also be generated. The user address generation algorithm consists of generating a key-pair and then extracting the public key (64-byte byte array representing x, y coordinates). Hash the public key using the SHA3-256 function (the SHA3 protocol adopted is KECCAK-256) and extract the last 20 bytes of the result. Add 41 to the beginning of the byte array and ensure the initial address length is 21 bytes. Hash the address twice using SHA3-256 function and take the first 4 bytes as verification code. Add the verification code to the end of the initial address and obtain the address in base58check format through base58 encoding. An encoded Mainnet address begins with T and is 34 bytes in length.

04

Account Data Structure

```
message Account {
        message Vote {
                bytes vote_address = 1;
                int64 vote_address = 2;
        }
        bytes account_name = 1;
        AccountType type = 2;
        byte address = 3;
        int64 balance = 4;
        repeated Vote votes = 5;
        map<string, in64> asset = 6;
        int64 latest_operation_time = 10;
}

enum AccountType {
        Normal = 0;
        AssetIssue = 1;
        Contract = 2;
}
```

4.1 QuarryChain Block Data Structure

```
Quarrychain.prototype.createNewBlock = function(none, previousBlockHash, hash) {
        const newBlock = {
                index: this.chain.length + 1,
                timestamp: Date.now(),
                transactions: this.pendingTransactions,
                nonce: nonce,
                hash: hash,
                previousBlockHash: previousBlockHash
        };

        this.pendingTransactions = [ ];
        this.chain.push(newBlock);

        return newBlock;
}
```

4.1.1 Hash Block Data Structure

```
Quarrychain.prototype.hashBlock = function(previousBlockHash, currentBlockData, nonce) {
        const dataAsString = previousBlockHash + nonce.toString() + JSON, stringify(currentBlockData);
        const hash = sha256(dataAsString);
        return hash;
};
```

4.1.2 Raw Data

Raw data is denoted as raw_data in Protobuf. It contains the raw data of a message, containing 6 parameters:

timestamp: Timestamp of this message -e.g. 1543884429000

txTrieRott: The Merkle Tree's Root -e.g. 7dasca...3ed

parentHash: The hash of the last block - e.g. 7dasca...3ed

number: The block height - e.g. 4638708

version: Reserved - e.g. 5

witness_address: The address of the witness packed in this block - e.g. 41928c...4d21

4.1.3 Witness Signature

Witness signature is denoted as witness_signature in Protobuf, which is the signature for this block header from the witness node.

4.1.4 Block ID

Block ID is denoted as blockID in Protobuf. It contains the atomic identification of a block. A Block ID contains 2 parameters:

hash: The hash of the block

number: The hash and height of the block

4.2 Transaction

Transaction data structure:

```
Quarrychain.prototype.createNewTransaction = function(amount, sender, recipient) {
        const newTransaction = {
                amount: amount,
                sender: sender,
                recipient: recipient,
        };

        this.pendingTransactions.push(newTransaction);

        return this.getLastBlock() ['index'] + 1;
};
```

4.2.1 Signing

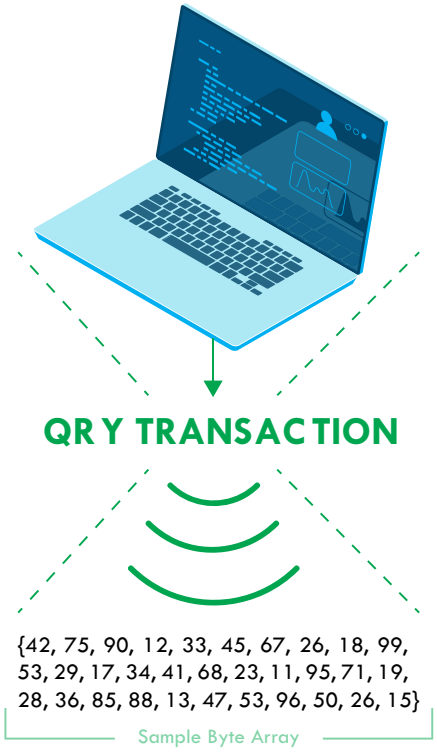
QuarryChain's transaction signing process follows a standard ECDSA cryptographic algorithm, with a SECP256K1 selection curve. A private key is a random number, and the public key is a point on the elliptic curve. The public key generation process consists of first generating a random number as a private key, and then multiplying the base point of the elliptic curve by the private key to obtain the public key. When a transaction occurs, the transaction raw data is first converted into byte format. The raw data then undergoes SHA-256 hashing. The private key corresponding to the contract address then signs the result of the SHA-256 hash. The signature result is then added to the transaction.

4.2.1 Signing

Ordinary transactions only consume bandwidth points, but smart contract operations consume both energy and bandwidth points. There are two types of bandwidth points available. Users can gain bandwidth points from freezing QRC, while 5000 free bandwidth points are also available daily.

When a QRC transaction is broadcast, it is transmitted and stored in the form of a byte array over the network. Bandwidth Points consumed by one transaction = number of transaction bytes multiplied by bandwidth points rate. For example, if the byte array length of a transaction

is 200, then the transaction consumes 200 bandwidth points. However, if a QRC or token transfer results in the target account being created, then only the bandwidth points consumed to create the account will be deducted, and additional bandwidth points will not be deducted. In an account creation scenario, the network will first consume the bandwidth points that the transaction initiator gained from freezing QRC. If this amount is insufficient, then the network consumes the transaction initiators QRC.



TRANSACTION BANDWIDTH CONSUMPTION

=

# of transaction bytes X bandwidth points rate

In standard QRC transfer scenarios from one QRC account to another, the network first consumes the bandwidth points gained by the transaction initiator for freezing QRC. If that is insufficient, it then consumes from the free 5000 daily bandwidth points. If that is still not enough, then the network consumes the QRC of the transaction initiator. The amount is calculated by the number of bytes in the transaction multiplied by 10 QRC. Thus, for most QRC holders who may not necessarily freeze their QRC to participate in QM voting, the first step is automatically skipped(since QRC balance frozen = 0) and the 5000 daily free bandwidth powers the transaction.

For the QRC-10 token transfers, the network first verifies whether the total free bandwidth points of the issued token asset are sufficient. If not, the bandwidth points obtained from freezing QRC are consumed. If there is still not enough bandwidth points, then it consumes the QRC of the transaction initiator.

4.2.3 Network Fee

The QuarryChain network generally does not charge fees for most transactions, however, due to system restrictions and fairness, bandwidth usage and transactions do take in certain fees.

Fee charges are broken down into the following categories:

1. Normal transactions cost bandwidth points. Users can use the free daily bandwidth points (5000) or freeze QRC to obtain more. When bandwidth points are not enough, QRC will be used directly from the sending account.

2. Smart contracts cost energy, but will also need bandwidth points for the transaction to be broadcast and confirmed. The bandwidth cost is the same as above. 3. All query transactions are free. It doesn't cost energy or bandwidth.

The QuarryChain Network also defines a set of fixed fees for the following transactions:

Creating a witness node..... 32 QRC

Issuing a QRC-10 token.....3 QRC

Creating a new account.....0.1 QRC

Creating an exchange pair..... 3 QRC

4.2.4 Transaction as Proof of Stake (TaPoS)

Quarrychain uses TaPoS to ensure the transactions all confirm the main blockchain, while making it difficult to forge counterfeit chains. In TaPoS, the networks require each transaction to include part of the hash of a recent block header. This requirement prevents transactions from being replayed on forks not including the referenced block, and also signals the network that a particular user and their stake are on a specific fork. This consensus mechanism protects the network against Denial of Service, 51%, selfish mining, and double spend attacks.

4.2.4 Transaction Confirmation

A transaction is included in a future block after being broadcast to the network. After 19 blocks are mined on Quarrychain (including its own block), the transaction is confirmed. Each block is produced by one of the top 27 Quarry Miner in a round robin fashion. Each block takes ~3 seconds to be mined on the blockchain. Time may slightly vary for each Quarry Miner due to network conditions and machine configurations. In general, a transaction is considered fully confirmed after ~1 minute.

4.2.6 API Transaction Structure

Transaction APIs consist of the following functions:

```
message Transaction {
        message Contract {
                enum ContractType {
                        AccountCreateContract = 0; // Create account/wallet
                        TransferContract = 1; // Transfer QRC
                        TransferAssetContract = 2; // Transfer QRC10 token
                        VoteWitnessContract = 4; // Vote for Super Representative (QM)
                        WitnessCreateContract = 5; // Create new QM account
                        AssetIssueContract = 6; // Create new QRC10 token
                        WitnessUpdateContract = 8; // Update QM information
                        ParticipateAssetIssueContract = 9; // Purchase QRC10 token
                        AccountUpdateContract = 10; // Update account/wallet information
                        FreezeBalanceContract = 11; // Freeze QRC for bandwidth or energy
                        UnfreezeBalanceContract = 12; // Unfreeze QRC
                        WithdrawalBalanceContract = 13; // Withdraw QM rewards, once per day
                        UnfreezeAssetContract = 14; // Unfreeze QRC10 token
                        UpdateAssetContract = 15; // Update a QRC10 token's information
                        ProposalCreateContract = 16; // Create a network proposal by any new QM
                        ProposalApproveContract = 17; // QM votes yes for a network proposal
                        ProposalDeleteContract = 18; // Delete a network proposal by owner
                        CreateSmartContract = 30; // Deploy a new smart contract
                        TriggerSmartContract = 31; // Call a function on a smart contract
                        GetContract = 32; // Get an existing smart contract
                        UpdateSettingContract = 33; // Update a smart contract's parameters
                        ExchangeCreateContract = 41; // Create a token trading pair on DEX
                        Exchange Inject Contract = 42; // Inject funding into a trading pair
                        ExchangeWithdrawContract = 43; // Withdraw funding from a trading pair
                        ExchangeTransactionContract = 44; // Perform token trading
                        UpdateEnergyLimitContract = 45; // Update orgin_energy_limit on a smart contract
                }
            }
        }
```

05

QuarryChain Virtual Machine

5.1 Introduction

QuarryChain's Virtual Machine (QVM) is a lightweight, Turing complete virtual machine developed for QuarryChain's Ecosystem. Its goal is to provide a custom-built blockchain system that is efficient, convenient, stable, secure and scalable.

QVM initially forked from EVM and can connect seamlessly with the existing solidity smart construct development ecosystem. Based on that, QVM additionally supports DPoS consensus.

QVM employs the concept of Energy. Different from the Gas mechanism on EVM, operations of transactions and smart contracts on QVM are free, with no QRC consumed. Technically, executable computation capacity on QVM is not restricted by the total holding amount of tokens.

5.2 Workflow

The compiler first translates the Solidity smart contract into bytecode readable and executable on the QVM. The QVM then processes data through opcode, which is equivalent to operating the logic of a stack-based finite state machine. Finally, the QVM accesses blockchain data and invokes External Data Interface through the Interoperation layer.

PERFORMANCE

Lightweight Architecture

Robust

High Compatibility

Low Cost

5.3.1 Lightweight Architecture

QVM adopts a lightweight architecture with the aim of reducing resource consumption to guarantee system performance.

5.3.2 Robust

QRC transfers and smart contract execution cost bandwidth points only, instead of QRC, which exempts the Quarrychain QVM is compatible with EVM and will be compatible with more mainstream VMs in the future. Thereby, all smart contracts on EVM are executable on QVM Chain Network from being attacked. Bandwidth consumption is predictable and static since each computational step cost is fixed.

5.3.3 High Compatibility

QVM is compatible with EVM and will be compatible with more mainstream VMs in the future. Thereby, all smart contracts on EVM are executable on QVM.

5.3.4 Low Cost

Due to QVM's bandwidth setup, development costs are reduced and developers can focus on the logic development of their contract code. QVM also offers all-in-one interfaces for contract deployment, triggering and viewing to offer the convenience for developers.

06

Smart Contract

6.1 Introduction

A smart contract is a protocol that digitally verifies contract negotiation. They define the rules and penalties related to an agreement and also automatically enforce those obligations. The smart contract code facilitates, verifies, and enforces the negotiation or performance of an agreement or transaction. From a tokenization perspective, smart contracts also facilitate automatic funds transfers between participating parties should certain criteria be met.

6.2 Energy Model

The maximum energy limit for deploying and triggering a smart contract is a function of several variables:

- Dynamic energy from freezing 1 QRC is 50,000,000,000 (Total Energy Limit) / (Total Energy Weight)
- Energy limit is the daily account energy limit from freezing QRC
- Remaining daily account energy from freezing QRC is calculated as Energy Limit - Energy Used
- Fee limit in QRC is set in smart contract deploy/trigger call
- Remaining usable QRC in the account
- Energy per QRC if purchased directly (10 Quarry = 1 Energy) = 100,000, QM can vote adjustment

There are two consumption scenarios to calculate the maximum energy limit for deployment and trigger. The logic can be expressed as follows:

```
const R = Dynamic Energy Limit
const F = Daily account energy from freezing QRC
const E = Remaining daily account energy from freezing QRC
const L = Fee limit in QRC set in deploy/trigger call
const T = Remaining usable QRC in account_name
const C = Energy per QRC if purchased directly

// Calculate M, defined as maximum energy limit for deployment/trigger of smart contract
if F > L*R
        let M - min(E+T*C, L*R)
else
        let M = E+T*C
```

6.3 Deployment

When a QuarryChain solidity smart contract is compiled, the Quarry Virtual Machine reads the compiled bytecode. The bytecode consists of a section for code deployment, contract code, and the Auxdata. The Auxdata is the source code's cryptographic fingerprint, used for verification. The deployment bytecode runs the constructor function and sets up the initial storage variables. The deployment code also calculates the contract code and returns it to the QVM. The ABI is a JSON file that describes the QuarryChain smart contract's functions. This file defines the function names, their payability, the function return values, and their state mutability.

6.4 Trigger Function

Once the QuarryChain's smart contracts are deployed, their functions can be triggered individually either via QuarryStudio or through API calls. State-changing functions require Energy while read-only functions execute without Energy.

6.5 QuarryChain Solidity

QuarryChain Solidity is a fork from Ethereum's Solidity language. The rest of the language syntax is compatible with Solidity ^0.4.24. Therefore the Quarry Virtual Machine (QVM) is almost 100% compatible with EVM instructions.

07

Token

7.1 QRC-10 Token

In the QuarryChain Network, each account can issue tokens at the expense of 3 QRC. To issue tokens, the issuer needs to specify a token name, the total capitalization, the exchange rate to QRC, circulation duration, description, website, maximum bandwidth consumption per account, total bandwidth consumption, and the amount of token frozen. Each token issuance can also configure each account's maximum daily token transfer Bandwidth Points, the entire network's maximum daily token transfer Bandwidth Points, total token supply, locking duration in days, and the total amount of tokens locked.

7.1 QRC-20 Token

QRC-20 Token is a technical standard used for smart contracts implementing tokens supported by the Quarry Virtual Machine. It is fully compatible with ERC-20.

The interface is as follows:

```
contract QRC20Interface {
        function totalsupply() public constant returns (uint);
        function balanceof(address tokenOwner) public constant returns (uint balance);
        function allowance(address tokenOwner, address spender) public constant returns (uint remaining);
        function transfer(address to, uint tokens) public returns (bool success);
        function approve(address spender, unit tokens) public returns (bool success);
        function transferFrom(address from, address to, uint tokens) public returns (bool success);

        event Transfer(address indexed from, address indexed to, uint tokens);
        event Approval(address indexed tokenOwner, address indexed spender, uint tokens);
}
```

From a developers perspective, there are several differences between QRC-10 and QRC-20. Some of the key differences are that QRC-10 tokens are accessible by APIs and smart contracts while QRC-20 tokens allow for interface customization but are only accessible within smart contracts. From a cost perspective QRC-10 tokens have transaction fees that are 1000 times lower than QRC-20, but carry bandwidth costs for API transfers and deposits. Transfers and deposits in smart contracts for QRC-10 tokens cost both bandwidth and energy.

7.3 Beyond

Since QuarryChain uses the same Solidity version as Ethereum, more token standards could be readily ported to QuarryChain.

08

Governance

Quarry Mining

8.1.1 General

The compiler first translates the Solidity smart contract into bytecode readable and executable on the QVM. Every account in the QuarryChain Network can apply and have the opportunity to become a Quarry Miner (denoted as QM). Everyone can vote for QM candidates. The top 20-30 candidates with the most votes will become QM with the right and obligation to generate blocks. The votes are counted every 6 hours and the QM will change accordingly.

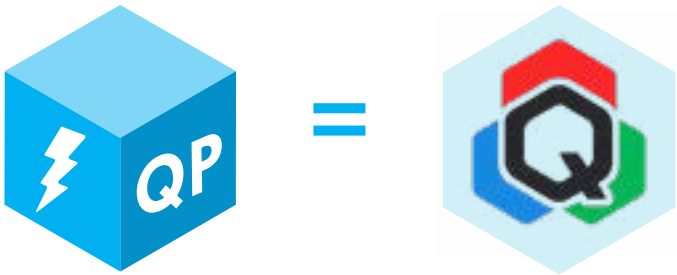
To prevent malicious attacks, there is a cost to becoming an QM candidate. When applying, 9999 QRC will be burned from the applicant's account. Once successful, such accounts can join the QM election.

The QVM then processes data through opcode, which is equivalent to operating the logic of a stack-based finite state machine. Finally, the QVM accesses blockchain data and invokes External Data Interface through the Interoperation layer.

8.1.1 General

Quarry Power (denoted as QP) is needed to vote and the amount of QP depends on the voter's frozen assets (QRC)

QP is calculated in the following way:



1 QuarryPower = 1 QRY Frozen to get bandwidth

Every account in the Quarrychain Network has the right to vote for their own QM.

After the release (unfreeze, available after 3 days), users won't have any frozen assets and lose all QP accordingly. As a result, all votes become invalid for the ongoing and future voting round unless QRC is frozen again to vote.

Note that the Quarrychain Network only records the most recent vote, which means that every new vote will negate all previous votes.

8.1.3 Reward

A. Vote Reward

Also known as a Candidate Reward, which the top 127 candidates updated once every round (6 hours) will share 230 QRC as mined. The reward will be split in accordance with the vote weight each candidate receives. Each year, the total reward for candidates will be 336,700 QRC.

Total Vote Reward per round

Why 230 QRC every round?

230 QRC = total value reward per round (V R/round)

V R/round = 0.03 QRC/block x 20 blocks/min x 60 mins/hr x 6hrs/round

(Note: this is set by WITNESS_STANDBY_ALLOWANCE = 230 QRC. See dynamic network parameters.)

Total vote reward per year

Why 336,700 QRC every year?

336,700 QRC = total vote reward per year (V R/year)

V R/year = 230 QRC/round x 4 rounds/day x 365 days/year

B. Block Reward

Also known as Quarry Miner Reward, which the top 27 candidates (QM) who are elected every round (6 hours) will share roughly 230 QRC as mined. The reward will be split evenly between the 27 QM (minus the total reward blocks missed due to network error). A total of 336,700 QRC will be awarded annually to the 27 QM.

Total block reward per round

Why 230 QRC every round?

230 QRC = total block reward per round (B R/round)

B R/round = 230 QRC/round x 4 rounds/day x 365 days/year

Total block reward per year

Why 336,700 QRY every year?

336,700 QRC = total block reward per year (B R/year)

B R/year = 230 QRY/round x 4 rounds/day x 365 days/year

January 1, 2022

There will be no inflation on the Quarrychain Network before January 1st, 2022, and the Quarrychain Foundation will award all block rewards and candidate rewards prior to that date.

C. Reward Calculation

QM reward calculation

total reward = vote reward (V R) + block reward (B R)

V R = total V R x votes QM candidates received / total votes

B R = total B R / 27 - block missed x 32

(Note: the reward is calculated per QM per round (6 hours))

Rank 28 to rank 127 QM candidate reward calculation

total reward = vote reward (V R)

V R = total V R x votes QM candidate received/total votes

(Note: the reward is calculated per QM candidate per round (6 hours))

COMMITTEE

8.2.1 General

The committee is used to modify the QuarryChain dynamic network parameters, such as block generation rewards, transaction fees, etc. The committee consists of the 27 QM in the current round. Each QM has the right to propose and vote on proposals. When a proposal receives 19 votes or more, it is approved and the new network parameters will be applied in the next maintenance period (3 days).

8.2.2 Proposal

Only the QM accounts have the rights to propose a change in dynamic network parameters.

8.2.3 Vote Proposal

Only committee members (QM) can vote for a proposal and the member who does not vote in time will be considered as a disagreement. The proposal is active for 3 days after it is created. The vote can be changed or retrieved during the 3-days voting window. Once the period ends, the proposal will either succeed (19+ votes) or fail (and end).

8.2.4 Cancel Proposal

The proposer can cancel the proposal before it becomes effective.

8.3 Structure

Only committee members (QM) can vote for a proposal and the member who does not vote in time will be considered as a disagreement. The pro QM are the witnesses of newly generated blocks. A witness contains 8 parameters:

1. **address**: the address of this witness - e.g. 0xuy62...7356
2. **voteCount**: number of received votes on this witness - e.g. 234234
3. **pubKey**: the public key for this witness - e.g. 0xuy62...7356
4. **url**: the url for this witness - e.g. https://www.noonetrust.com
5. **totalProduced**: the number of blocks this witness produced - e.g. 2434
6. **totalMissed**: the number of blocks this witness missed - e.g. 7
7. **latestBlockNum**: the latest height of block -e.g. 4522
8. **isjobs**: a boolean flag

Protobuf Data Structure:

```
message Witness{
        bytes address = 1;
        int64 voteCount = 2;
        bytes pubKey = 3;
        string url = 4;
        int64 totalProduced = 5;
        int64 totalMissed = 6;
        int64 latestBlockNum = 7;
        bool isJobs = 8;
}
```

09

DApp Development

9.1 APIs

The QuarryChain Network offers a wide selection of over 60+ HTTP API gateways for interacting with the network via Fill and Solidity Nodes. Additionally, QuarryWeb is a comprehensive JavaScript Library containing API functions that enable developers to deploy smart contracts, change the blockchain state, query blockchain and contract information, trade on the DEX, and much more. These API gateways can be directed towards a local privatenet, the Shasta testnet, or the QuarryChain Mainnet.

HTTP GATEWAYS USING FILL & SOLIDITY NODES

9.2 Networks

QuarryChain has both a Shasta testnet as well as a Mainnet. Developers may connect to the networks by deploying nodes, interacting via QuarryStudio, or using APIs via the QuarryGrid service. The QuarryGrid service consists of load balanced node clusters hosted on AWS servers worldwide. As DApp development scales up and API call volumes increase, QuarryGrid successfully fields the increase in API traffic.

9.3 Tools

QuarryChain offers a suite of development tools for enabling developers to create innovative DApps. QuarryBox is a framework that allows developers to test and deploy smart contracts via the QuarryWeb API. QuarryGrid is a load balanced and hosted API service that allows developers to access the QuarryChain network without having to run their own node. QuarryGrid offers access to both the Shasta testnet as well as the QuarryChain Mainnet. QuarryStudio is a comprehensive Integrated Development Environment (IDE) that enables developers to compile, deploy, and debug their Solidity smart contracts. QuarryStudio contains an internal full node that creates a private local environment for smart contract testing prior to deployment. The QuarryWeb API library connects developers to the network via a wide selection of HTTP API calls wrapped in JavaScript.

9.4 Resources

The QuarryChain Developer Hub is a comprehensive API documentation site tailored towards developers wishing to build on the QuarryChain Network. The Developer Hub provides a high-level conceptual understanding of QuarryChain and walks users through the details of interacting with the network. The guides walk developers through node setup, deployment and interaction with smart contracts, API interaction and implementation, building sample DApps, and using each of the developer tools. Additionally, developer community channels are available through Discord.

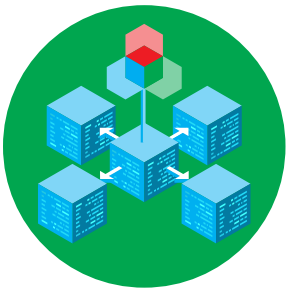
10

The QuarryChain Ecosystem

10.1 Introduction

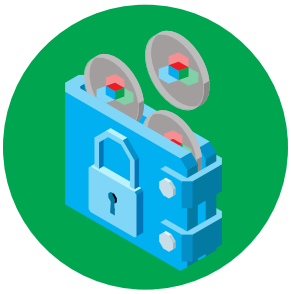
QuarryChain is a robust blockchain ecosystem designed and developed by the QuarryLabs Development Team There are multiple kinds of products involved in QuarryChain's Ecosystem, including public chain, wallet client, and decentralized applications, etc. These products are closely related to each other, and together, this supports the stability of the whole ecosystem.

10.2 Core Applications in QuarryChain Ecosystem



1. Overview

QuarryChain's public chain is a large-scale distributed operating system, having tens of thousands of nodes running on service, and terminals all over the world. Like Office to Microsoft, the power of QuarryChain's public chain cannot be separated from the support of application software, and the prosperity of QuarryChain's Ecosystem needs community developers to work together.



2. Wallet Client

QuarryWallet-Cli is the official wallet client maintained by the QuarryChain Foundation.

QuarryWallet-Cli is a command-line version of the wallet, provides essential tools to communicate with the Quarry public chain by the RPC protocol, supports all functions of the Quarry public chain in real-time.

QuarryWallet-Cli is also an excellent code implementation to show the interactive mode between clients and the blockchain for developers.

2.1
Quarry Wallet

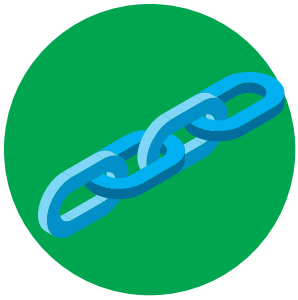
Quarry Wallet is an outstanding Digital Wallet dedicated to providing users with complete functions, convenient experience, secure funding options, and the most applications.

3.
Blockchain Exlporer

quarryscan.org is the first blockchain explorer based on QuarryChain.

quarryscan.org has all the essential functions of a blockchain explorer, including searching transactions/ accounts/ blocks/ nodes/ smart contracts, on-chain statistics, token creation, Etc.

quarryscan.org also has a built-in web wallet and a DEX based on the Bancor Protocol, and they enrich the application matrix of Quarryscan.



4.
DApp

QuarryChain's public chain, plans on attracting tens of thousands of community developers to join in the development, deployment, and running of DApps on the blockchain, for the reason of its high performance, safety, and low cost.

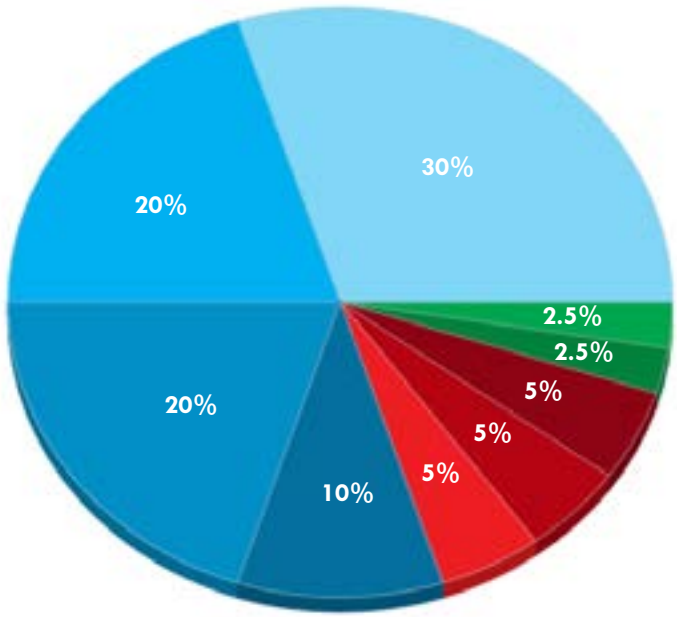
5.
Testnet

Shashta test net

QUARRY (QRY)

Quarry (QRY) is the official cryptocurrency and utility token for the QuarryChain Network. Quarry aims to be a peer-to-peer digital currency that will enable lightning fast low cost transactions to anyone in the world. Quarry will also be used to access and connect the entire QuarryChain Ecosystem, with an abundance of application scenarios that power transactions and applications on the chain.

10.3.1 Quarry Tokenomics



TOKEN ALLOCATION

- PRIVATE PRESALE
- PRIVATE SALE
- ANGEL INVESTORS
- MARKETING
- ECOSYSTEM
- DEV
- TEAM
- STAKING & FARMING
- PUBLIC SALE / LIQUIDITY

Total Supply	2.00E + 8 (200,000,000)
Decimals	18
Symbol	QRC
Name	Quarry Coin
Funding Goal	$50,000,000
Suggested Base Price	$0.0500000
Base Price	$0.100000

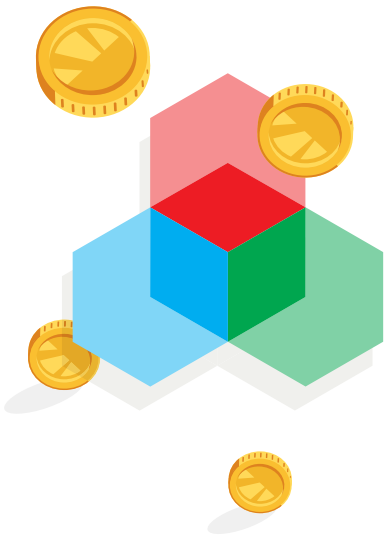
10.3.2 Funds Usage

35% of funds will be used to build the QuarryChain Platform, and this percentage will also be used for system upgrades. These upgrades will include team recruiting, training, and the development budget.

50% of the funds will be used for QuarryChain's Branding and Marketing, that'll include continuous promotions and education of QuarryChain's Blockchain Industry Innovations. QuarryChain aims to use its advertising to become popular among Cryptocurrency Investors as well as attract active users to the platform.

15% will be kept in reserve to cope with any emergency or in the case of an unexpected situation that might come up.

TOKENIZING REAL WORLD ASSETS



10.4.1 Introduction

Today the most common application of tokens that exist on DLT/Blockchains are tokens that are fungible of nature, like cryptocurrencies or ingots of gold. Fungible means that each unit of a good or commodity is interchangeable with any other unit. So, for example one kilo of pure gold is interchangeable with any other kilo of pure gold.

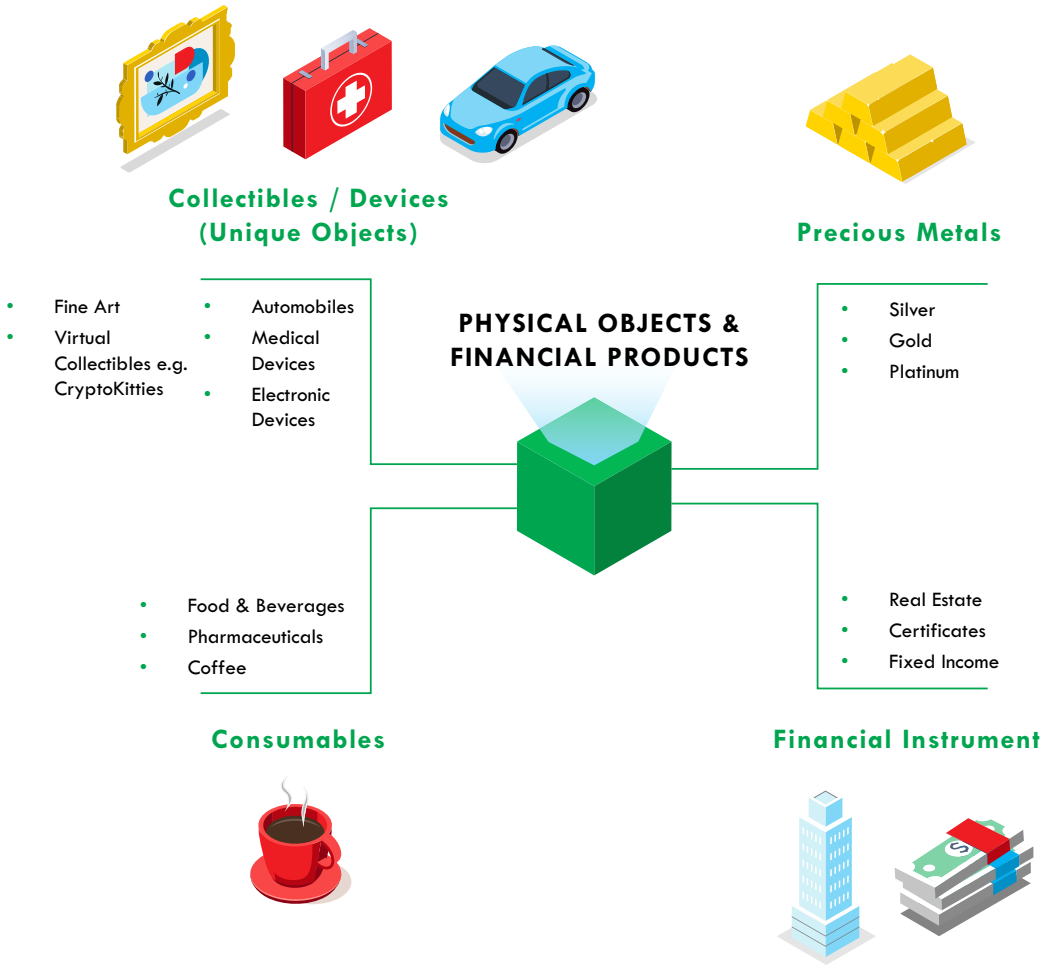
However, a new application concept of Blockchain Technology is causing a lot of buzz and gaining momentum: the tokenization of real world assets.

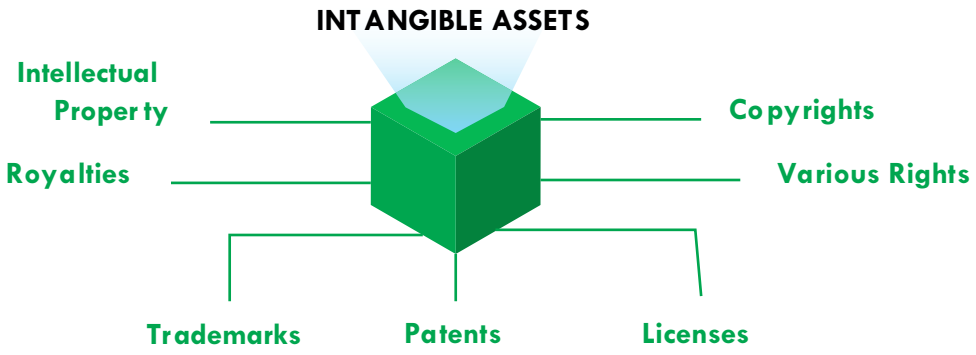
In the most abstract form, tokenization converts the value stored in a tangible or intangible object into a token that usually can be manipulated along a DLT / Blockchain system. In simple words, tokenization can turn almost any asset, either real or virtual, into a digital token and enables the digital transfer, ownership and storage without the necessary need of a central third party / Intermediary.

A digital token can thereby be described as a piece of software with a unique asset reference, properties and/or legal rights attached.

10.4.2 Tokenizing Industries

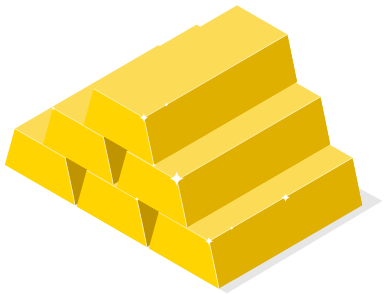
The QuarryChain Network is creating a solution that would tokenize multiple industries and create unimaginable liquidity across the whole global assets market. The process of tokenization will create a bridge between real-world assets and their trading, storage and transfer in a digital world.





10.4.4 Fungible Goods

The next layer of innovation arrives with fungible goods. A good is fungible when it can be exchanged for another identical good of equal value. The most familiar fungible goods are commodities. A liter of water is equal to another liter of water, as a barrel of oil is equal to another barrel of oil or an ounce of gold is equal to another ounce of gold. Even stocks can be considered to be fungible, provided they are grouped together in identical packages. Very often, fungible assets are backed by a physical resource, somewhere – gold or wheat in a warehouse, water or oil in a pipeline, etc.



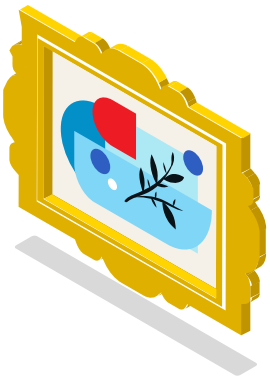
This property makes them difficult to physically trade. The difficulty is compounded when the scale of transactions comes into play. Fungible assets are often dealt with in bulk form, and delivery simply cannot be done instantaneously. A shipment of 10,000 short tons of line pipe, for instance, is pretty bulky. Transferring ownership of that asset from one entity to another either involves moving 10,000 short tons of steel or creating a paper trail, whereby the steel is transferred via a trusted third party, like a bank, to the new owner before it physically moves.

QuarryChain's Tokenization cuts much of the work out of this process. A digital representation of

steel, for instance, can be traded between two parties on QuarryChain utilizing QRC-20 Smart Contracts. There are no intermediaries in this process - no exchange agents, no port officials, government checks, or warehouses. The steel, which is uniquely identified on QuarryChain, is moved from the buyer and seller instantaneously, along with auxiliary shipping or warehouse information. The sale is then recorded on QuarryChain so as to form a permanent and instantly verifiable receipt. This replaces the traditional paper record-keeping system and enables the exchange of fungible goods on a much more detailed and precise scale.

10.4.4 Non-Fungible Goods

Here's where blockchain technology really gets interesting. Tokenization enables real world, non-fungible goods to be parceled out into digital "shares", which can then be bought, sold, or traded in a full or limited fashion with the public. The two most compelling use cases offered so far concern art and real estate. For example, in the whole world there can only be one real Mona Lisa painting. This physical painting is one of a kind, and it can only be bought or sold as a unit - that is, the smallest divisible unit of the Mona Lisa is one Mona Lisa. Moreover, this Mona Lisa is not the same as the millions of prints or digital copies of the Mona Lisa. In other words, a picture or poster of the Mona Lisa is not the Mona Lisa itself, and it doesn't carry the same value.



Tokenizing a work of art introduces a digital signature that cannot be altered. The digital token representing the Mona Lisa is one of a kind. It is not a copy, but the token can be broken down into sub-tokens, each also digitally signed. In this way, "shares" of a unique piece of art can be sold to the general public. The same goes for unique pieces of real estate.

The ability to tokenize unique, non-fungible assets means that ownership can be distributed. Funds can be raised more easily, and a broader group of entities takes responsibility for the care and upkeep of that item. Each holder of a Mona Lisa token doesn't have a copy of the Mona Lisa, they actually own a part of the artwork itself, which they can keep as a store of value or sell to another willing buyer.

10.4.5 Technical Approach to Tokenize Assets

Blockchain Technology allows the creation of a digital representation of assets through their tokenization. This process does not greatly differ from other IT projects whereby data models are created to represent assets or goods.

In simple terms, asset tokenization consists of creating an informatic code presenting the key characteristics of the assets while exposing some functions allowing the user to interact with the digital representation of the asset. On QuarryChain, this informatic code is developed in Solidity.

From a technical standpoint, this process can be broken down into 4 key steps:

1. SELECTION OF THE MODEL REPRESENTING ASSETS

Several token standards have emerged from the QuarryChain protocol to allow for the representation of different kinds of assets. The introduction of these standards eases the Blockchain adoption and enables interoperability between multiple DLT/Blockchain initiatives.

This allows tokens inheriting from the same token standard (from the same 'family') to use smart contracts or to be stored in widespread wallets enabling the storage of ownership information.

A token standard consists of a set of predefined functions and/or attributes that can be intentionally implemented to represent the specificities of each asset but need to be present in all cases. To consider which token standard to choose, the key characteristics of the asset (for instance fungible or non-fungible) should be assessed. Here are the following Token Standards available for the QuarryChain Network:

QRC-20 Standard
Fungible Tokens (e.g. Ether, Augur, etc)

- Transfer of value between users
- Authorize someone to spend value on someone's behalf

QRC-1400 Standard
Security tokens (e.g. data protection & privacy)

- Transfer of ownership of a security token between users, requiring a certificate
- For same security token, split between several partitions
- Authorize an operator to transfer a security token on one's behalf
- Manage the documentation associated with a security token
- Force the transfer or redemption of an asset (by a controller)

QRC-721 Standard
Non-fungible tokens (e.g. Crypto Kitties, assets)

- Transfer of ownership over a specific asset between users
- Authorize an operator to transfer a specific asset on one's behalf
- Authorize an operator to transfer all assets on one's behalf

2. MODELLING OF THE ASSET

Before implementing the token model chosen for representing assets, one must ask several questions that will impact which information will be embedded on chain and which will remain stored in off-chain databases. The following points should at least be considered:

- Are there any legal or regulatory constraints (e.g. data privacy and protection, sector-specific regulation, etc.)?
- What is the level of trust required on the data?
- What is the business process? Which information is essential for the process to happen properly and which has an informative purpose?
- What are the requirements in terms of scalability (volumetry of data)?

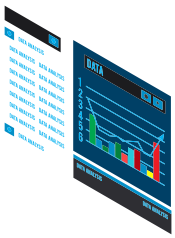
This tailoring can include the management of permissioning to call specific functions, adding some additional functions to be consistent with business requirements, the personalization of some generic functions, etc. Some additional considerations may be needed to develop the specific code and behavior of the token, depending on the kind of assets considered. For instance, if financial assets are considered, their issuance, the management of the liquidity or the way they will be valued on both primary and secondary markets should be properly engineered.

It is also important to note that the documentation of the behavior of the token is key for ensuring the adoption and the use of this asset: especially in the case of the issuance of financial assets, a term sheet is frequently released to help investors and/or users understand the functioning and the underlying key concepts of the asset. This term sheet may also be required by regulators to approve the issuance of the digital asset, that is the case in some STO's for instance.

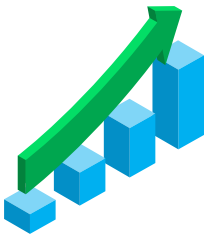
CONSIDERATIONS

Legalities & Regulations

Level of Trust Required



Business Processes & Data



Scalability & Growth

3. TECHNICAL & SECURITY OF THE INFORMATIC CODE

Before deploying a token on QuarryChain, some precautions need to be taken due to the immutability of Blockchain and the irreversibility of the development. Indeed, once the code ruling on how your asset will behave on DLT/Blockchain has been released, it is not possible to turn back.

An example to bear in mind is what happened in June 2016 with "The DAO" which was a digital decentralized autonomous organization raising funds through crowdfunding and acting as a venture capital. After raising more than $150 million, some hackers took advantage of vulnerability issues and stole more than the equivalent of $50 million at the time.

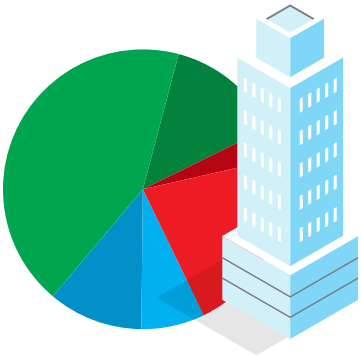
Therefore, it is very essential to perform automated security audits on QuarryChain, either utilitizing QuarryChain's Standardized Applications or using third party auditors (CERTIK, etc.) to perform an audit and provide a stamp before deploying the contract.

4. DEPLOYMENT OF THE INFORMATIC CODE

After the security audit has been performed, the code can then simply be deployed on QuarryChain and be considered, either public or private depending on the use case and perimeter considered.

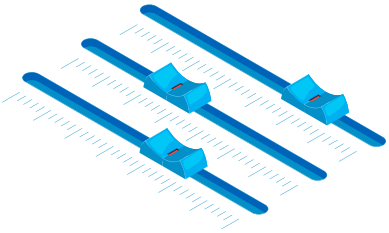
These tokens will be able to be issued following the design implement, automatically or manually, by only one user or by multiple ones. Users will be able to interact with these tokens using the exposed function, by instance to transfer them.

10.4.6 Benefits of Tokenization



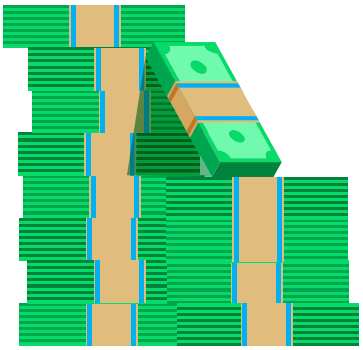
FRACTIONALIZATION

- Assets such as real estate have high barrier entry due to larger upfront capital required.
- Fractionalizing such assets democratizes its access for smaller investors.



CUSTOMIZABILITY

- Tokenization enables exposure to individual real estate assets.
- Thus, instead of investing in the whole sector, portfolios can be customized down to single buildings.



LIQUIDITY

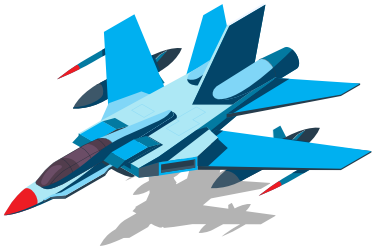
- Fractionalization increases the pool of potential investors and can unlock global investor base.
- Secondary markets also facilitate additional liquidity.
- Liquid assets command a premium and can increase asset value.

AUTOMATION

- QRC-20 Smart Contracts can automate steps such as compliance, document verification, trading, and escrow.
- Dividends and other cash flows can be programmatically paid when due.

COST EFFICIENCY

- By removing certain intermediaries and increasing efficiency of processes, costs can be lowered.

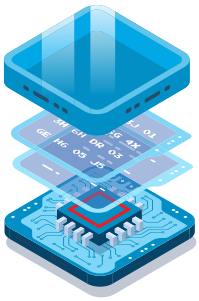


SETTLEMENT TIME

- QRC-20 Tokens can be settled within seconds through QuarryChain's DPoS consensus.
- This unlocks the capital that is tied in the market which currently settles at T+3/T+2.

DATA TRANSPARENCY

- Secure and visible record keeping on QuarryChain can increase transparency to the underlying data.



STRUCTURED PRODUCTS

- Additional value can be realized once assets are tokenized and that enables the creation of additional layered financial products such as basket of assets and derivatives.
- Since the underlying is tokenized, creating complex products becomes simpler through coded smart contracts.

11 QuarryChain Terms and Conditions

11.1 General Information

QuarryChain Ecosystem Tokens, do not have the legal qualifications of a security, since it does not give any rights to dividends or interests. QuarryChain Tokens are not shares and do not give any right to participate to the general meeting of QuarryLabs Inc. QuarryChain Tokens cannot have a performance or a particular value outside the QuarryLabs Inc network. QuarryChain Tokens shall therefore not be used or purchased for speculative or investment purposes. The purchaser of QuarryChain Tokens is aware that national securities laws, which ensure that investors are sold investments that include all the proper disclosures and are subject to regulatory scrutiny for the investors' protection, are not applicable. Anyone purchasing QuarryChain Tokens expressly acknowledges and represents that she/he has carefully reviewed this whitepaper and fully understands the risks, costs and benefits associated with the purchase of QuarryChain Tokens.

11.1 General Knowledge

The purchaser of QuarryChain Tokens undertakes that she/he understands and has significant experience of cryptocurrencies, blockchain systems and services, and that she/he fully understands the risks associated with the crowd sale as well as the mechanism related to the use of cryptocurrencies (incl. storage). QuarryLabs Inc shall not be responsible for any loss of QuarryChain Tokens or situations making it impossible to access QuarryChain Tokens, which may result from any actions or omissions of the user or any person undertaking to acquire QuarryChain Tokens, as well as in case of hacker attacks.

11.3 Risks

Acquiring QuarryChain Tokens and storing them involves various risks, in particular the risk that QuarryLabs Inc may not be able to launch its operations and develop its blockchain and provide the services promised. Therefore, and prior to acquiring QuarryChain Tokens, any user should carefully consider the risks, costs and benefits of acquiring QuarryChain Tokens in the context of the crowd sale and, if necessary, obtain any independent advice in this regard. Any interested person who is not in the position to accept or to understand the risks associated with the activity (incl. the risks related to the non-development of the QuarryChain platform) or any other risks as indicated in the Terms & Conditions of the crowd sale should not acquire QuarryChain Tokens.

11.3 Risks

This whitepaper shall not and cannot be considered as an invitation to enter into an investment. It does not constitute or relate in any way nor should be considered as an offering of securities in any jurisdiction. The whitepaper does not include nor contain any information or indication that might be considered as a recommendation or that might be used to base any investment decision. This document does not constitute an offer or an invitation to sell shares, securities or rights belonging to QuarryLabs Inc or any related or associated company. The QuarryChain Tokens are just utility tokens which can be used only on the QuarryChain platform and is not intended to be used as an investment.

The offering of QuarryChain Tokens on a trading platform is done in order to allow the use of the QuarryChain platform and not for speculative purposes. The offering of QuarryChain Tokens on a trading platform is not changing the legal qualification of the token, which remains a simple means for the use of the QuarryChain platform and is not a security.

QuarryLabs Inc is not to be considered as advisor in any legal, tax or financial matters. Any information in the whitepaper is given for general information purpose only and QuarryLabs Inc does not provide with any warranty as to the accuracy and completeness of this information. Given the lack of crypto-token qualifications in most countries, each buyer is strongly advised to carry out a legal and tax analysis concerning the purchase and ownership of QuarryChain's Tokens according to their nationality and place of residence. QuarryLabs Inc today is not a financial intermediary according to US Law and is not required to obtain any authorization for Anti-Money Laundering purpose. This qualification may change in case QuarryLabs Inc will offer services which are to be considered as qualifying a financial intermediation activity.

QuarryChain Tokens confer no direct or indirect right to QuarryLabs Inc capital or income, nor does it confer any governance right within QuarryLabs Inc. Regulatory authorities are carefully scrutinizing businesses and operations associated with cryptocurrencies in the world. In that respect, regulatory measures, investigations or actions may impact QuarryLabs Inc and even limit or prevent it from developing its operations in the future. Any person undertaking to acquire QuarryChain Tokens must be aware of the QuarryChain business model, the whitepaper or Terms & Conditions may change or need to be modified because of new regulatory and compliance requirements from any applicable laws in any jurisdictions. In such a case, purchasers and anyone undertaking to acquire QuarryChain Tokens acknowledge and understand that neither QuarryLabs Inc nor any of its affiliates shall be held liable for any direct or indirect loss or damage caused by such changes.

QuarryLabs Inc will do its utmost to launch its operations and develop the QuarryChain platform. Anyone undertaking to acquire QuarryChain Tokens acknowledges and understands that QuarryLabs Inc does not provide any guarantee that it will manage to achieve it. On concluding the Commercial Operation, these tokens will be issued by a technical process referred to as a «Blockchain». This is an open source IT protocol over which the Company has no rights or liability

in terms of its development and operation.

The token distribution mechanism will be controlled by a Smart Contract; this involves a computer program that can be executed on the QuarryChain network or on a blockchain network that is compatible with Smart Contract programming language. They acknowledge and understand therefore that QuarryLabs Inc (incl. its bodies and employees) assumes no liability or responsibility for any loss or damage that would result from or relate to the incapacity to use QuarryChain Tokens, except in case of intentional misconduct or gross negligence.

QuarryChain Tokens is based on the QuarryChain protocol. Therefore, any malfunction, unplanned function or unexpected operation of the Binance protocol may cause the QuarryChain network or QuarryLabs Inc to malfunction or operate in a way that is not expected. Ether, the native Binance Protocol account unit may itself lose value in a similar way to QuarryChain Tokens, and also in other ways.

12 Conclusion

QuarryChain is pioneering a new era in blockchain technology, where scalability, efficiency, and accessibility converge to create a network that meets the demands of a rapidly evolving digital landscape. Our mission is to overcome the limitations of traditional blockchain systems by providing a robust platform capable of handling an immense volume of transactions and supporting a vast user base.

Our goals extend beyond mere technological achievements. We are committed to fostering an inclusive and decentralized ecosystem where users have full autonomy over their digital assets and data. By delivering high-speed, low-cost transactions with an environmentally conscious approach, QuarryChain is set to become a foundational infrastructure for a wide range of industries, from finance and real estate to energy and health.

"Get Quality with Quarry" encapsulates our dedication to excellence in every aspect of our network. We believe that quality is the key to building trust and driving widespread adoption, and we are unwavering in our pursuit of providing a blockchain that is reliable, secure, and user-friendly.

As we continue to grow and evolve, our focus remains on innovation and scalability, ensuring that QuarryChain can adapt to the future needs of the global blockchain community. We are committed to making QuarryChain a leader in the decentralized world, enabling users everywhere to harness the full potential of blockchain technology.

13 Glossary

TERM	DEFINITION
Address / Wallet	An address or wallet consisting of account credentials on the QuarryChain network are generated by a key pair, which consists of a private key and a public key, the latter being derived from the former through an algorithm. The public key is Usually used for session key encryption, signature verification, and encrypting data that could be decrypted by a corresponding private key.
ABI	An application binary interface (ABI) is an interface between two binary program modules; usually one of these modules is a library or an operating system facility, and the other is a user run program.
Bandwidth Points (BP)	To keep the network operating smoothly, the QuarryChain network transactions use BP as fuel. Each account gets 5000 free daily BP and more can be obtained by freezing QRC for BP. Both QRC and QRC-10 token transfers are normal transactions costing BP. Smart contract deployment and execution transactions consume both BP and Energy.
Byzantine General's Problem	At a conceptual level, blockchain attempts to address the challenge posed by a paper published in 1975 that outlines what has become known as the Byzantine Generals Problem. Two generals commanding their own armies are attacking a common enemy. Neither army is powerful enough to defeat the enemy on its own, so the generals must send messengers to each other confirming the time of attack. General 1 sends a messenger with the specified time of attack to General 2. If the messenger is not captured and arrives, how does General 2 know that the messenger has not been compromised by the enemy? Even if the messenger has not been captured or compromised, how does General 1 know that the messenger returning with the confirmed time of attack has not been compromised? The Generals can never be sure that the information they are receiving is accurate and valid.

TERM	DEFINITION
Block	A list of recorded transactions over a period of time. Transactions can represent virtually any type of activity, from registering a land deed to a single purchase. Any rules relating to the block itself are established when the Network is first created.
Block Reward	Block production rewards are sent to a sub-account (address/wallet). Quarry Miner can claim their rewards on Quarryscan or through the API directly.
Block Header	A block header is part of a block. QuarryChain block headers contain the previous block hash, the Merkle root, timestamp, version, and witness address.
Consensus Mechanism	The newest blockchain must be consistent and accepted by the majority of computers online. The method by which the new blockchain is accepted is defined by the algorithm used.
Cryptocurrency	A digital currency that encryption techniques use to regulate the generation of units of currency and verify the transfer of funds, operating independently of a central bank.
Chain	When the Block reaches its maximum size of transactions, it is chained or linked to the proceeding block through a hash(described in the section below) The hash value of one Block is inserted into the next Block. This makes a link between the new Block and the previous Block. Repeating a hash function on an unaltered block of data will always generate the same fixed length value. If a Block of Data is altered, the resulting hash output will be different. A user can then see the hashes are different and will know the original Block has been altered and may no longer be trustworthy.
Decentralized Application (DApps)	Decentralized Application is an App that operates without a centrally trusted party. An application that enables direct interaction/agreements/ communication between end users and/or resources without a middleman.

TERM	DEFINITION
Digital ID	Information on an entity used by computer systems to represent an external agent. That agent may be a person, organization, application, or device.
Distributed Ledger	A replicated database shared across many nodes.
Hash	A function that takes an input string, which can be of any length, and generates an output of fixed length. The output, or hash, is used to authenticate information. The values returned by a hash function are called hash values, hash codes, digests, or simply hashes.
JDK	Java Development Kit is the Java SDK used for Java applications. It is the core of Java development, comprising the Java application environment (JVM+Java class library) and Java tools.
Merkle Tree	In cryptography, a Merkle tree is a tree in which every leaf node is labeled with the hash of a data block and every non-leaf node is labeled with the cryptographic hash of the labels of its child nodes. Hash trees allow efficient and secure verification of the contents of large data structures.
Mining	The process by which nodes compete to add transaction records or blocks to the blockchain.
Network	The Network is made up of nodes each containing a complete record of all transactions on a blockchain. No centralized “official” copy exists and no node is “trusted” more than another. The Data integrity is maintained by the blockchain being replicated on all of the nodes. The best way to think of a Node would be to think of it as a cluster of servers running on a Blockchain. Node operators are incentivized to operate on a node by receiving rewards for their efforts.

TERM	DEFINITION
Quarry Coin	Quarry Coin is the official cryptocurrency for the QuarryChain Protocol. Quarry Coin aims to be a peer-to-peer internet currency that enables instant low cost transactions to anyone in the world. Quarry Coin will feature faster transaction confirmation times and improved storage efficiency than the leading math-based currency.
Peer-to-Peer	A distributed network in which nodes are interconnected and share data directly with each other without a centralized service.
Permissionless / Public Ledger	Anyone can insert transactions into a block to be included in the blockchain.
Permissioned / Public Ledger	Restricted to a limited set of authorized nodes and only an authorized user can access, generate transactions, and update the ledger.
Public / Private Keys	Infrastructure Keys that are mathematically related to each other. The public key can be used to verify a signature generated using a private key.
QRC-10	A standard of crypto token on the QuarryChain Platform. Certain rules and interfaces are required to follow when holding an initial coin offering on the QuarryChain Network.
QRC-20	QRC-20 Token is a technical standard used for smart contracts implementing tokens supported by the Quarry Virtual Machine. It is fully compatible with ERC-20.
QRC	QRC stands for Quarry, which is the official cryptocurrency of QuarryChain.
QuarryChain Virtual Machine (QVM)	QuarryChain's Virtual Machine (QVM) is a lightweight, Turing complete virtual machine developed for QuarryChain's Ecosystem. Its goal is to provide a custom-built blockchain system that is efficient, convenient, stable, secure and scalable.
SHA-256	Secure Hash Algorithm (SHA) used to generate a 256-bit hash.

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